

Fundamentals of Data Science

MPhil in Economics and Data Science
Michaelmas Term 2024
Faculty of Economics
University of Cambridge

Course Information

- **Lecturers:**

- Dr. Adrian Ochs (acro2@cam.ac.uk)
- Dr. Christian Rörig (cmr61@cam.ac.uk)

- **Office Hours:**

- Dr. Adrian Ochs: [tbd]
- Dr. Christian Rörig: [tbd]

- **Lectures:** 9 lectures (2 hours each)

- **Classes:** 5 classes (2 hours each)

- **Teaching Assistant:** Dr. Niklas Schmitz (ns751@cam.ac.uk)

Course Overview

Description

This course introduces students to the fundamental concepts, techniques, and tools in data science. With a focus on end-to-end data science projects, the course is designed to equip students with the skills necessary for successful interviews and careers in the field. Students will learn to tackle real-world data problems, covering the entire spectrum from data acquisition and preprocessing to analysis, visualisation, statistical modelling, and considerations for moving models to production. As part of this course, students will also be equipped with valuable software engineering skills.

A note on course flexibility and structure

As this is a newly developed course, the outline may undergo minor adjustments throughout the semester to optimise your learning experience. Additional readings may be assigned to supplement the core material and problem sets. However, the primary focus will remain on practical hands-on learning experiences.

Outline

1. Lifecycle of a data science project
 - Introduction to typical stages of a data science project
2. Project setup phase
 - Initial project scoping and trade-offs
 - Setting up a data science repository

- Setup of pre-commit hooks, environment.yaml, .gitignore, etc.
- 3. Data wrangling and exploration
 - Hands-on data exploration sessions using Python's pandas and polars libraries
 - Coping with messy real-world data (e.g. missing values, outliers)
- 4. Data retrieval and storage
 - Data types and formats
 - Loading data files (e.g., CSV, Parquet)
 - Working with SQL databases
- 5. Data visualisation
 - Best practices for effective visualisations
 - Utilising libraries such as matplotlib, seaborn, plotly, and altair
- 6. Feature engineering
 - Feature engineering for numeric and categorical features
 - Dimension reduction techniques
 - FE pipelines using scikit-learn
- 7. Statistical modelling
 - Modeling strategy
 - Cross-Validation
 - Feature selection
 - Hyperparameter tuning
 - Advanced modeling techniques (e.g. model composition, custom loss function)
- 8. Model evaluation and interpretability
 - Evaluation metrics
 - Interpreting models using model-agnostic methods
- 9. Model tracking and reproducibility
 - Using MLflow for experiment tracking
 - Model conversion to ONNX format

Course Aims and Objectives

- You will be prepared for most data science internships/jobs after taking this class. In particular, you will be able to successfully complete a coding challenge end-to-end. These challenges are supposed to assess whether you have the fundamental DS skills to be successful in the company. So, this makes for a good guide on what 'fundamentals of DS' means.
- You will learn to start your own coding projects and write your own package, allowing you to develop a portfolio of projects on GitHub.
- You will learn to implement and review good coding practices.
- You will become familiar with the typical data science Python libraries (e.g. pandas, scikit-learn etc.).
- You will learn to collaborate on a common code base.

Course Materials

Available via Moodle and the course's GitHub repository [links will follow].

Computing Environment

Your laptop with

- VS Code as IDE
- Conda for package management
- Python as the primary programming language
- Git and GitHub for version control and collaboration

Classes

Students will be divided into groups for class assignments. You are expected to divide tasks and work collaboratively. This means that you will work with pull requests (PR) and PR reviews. We believe no one ever truly appreciates the value add of Git and GitHub until they worked in a team. Hence, part of the fundamentals of data science is collaboration including:

- How to divide tasks in a project.
- How to work collaboratively on code.
- How to maintain consistent code quality.

Assessment

The course will culminate in a final data science project, evaluating:

- Project structure and organization
- Code quality
- Data analysis methodology and approach
- A concise project report including visualization and presentation of results

Prior to the final project, students will gain hands-on exposure to all necessary tools and techniques.

Feedback

We welcome any feedback to improve this course. Please feel free to reach out via email or open an issue in the course's GitHub repository, even for minor suggestions such as typo corrections.